

netic waves generated by the reader to activate the tag and read/write data from it.

Antennae are the conduits between the tag and the transceiver which control the system's data acquisition and communication. These are available in a variety of shapes and sizes. Often, the antenna is packaged with the transceiver and decoder to become a reader, which can be configured either as a handheld or a fixed-mount device. The reader emits radio waves in ranges

of anywhere from 2.54 cm (one inch) to 30 metres or more, depending upon its power output and the radio frequency used. When an RFID tag passes through the electromagnetic zone, it detects the reader's activation signal. The reader decodes the data encoded in the tag's integrated circuit (silicon chip) and the data is passed to the host computer for processing.

Tags (transponders). Fig. 4 shows the internal structure of a typical RFID tag. An RFID tag comprises a micro-

chip containing identifying information and an antenna that transmits this data wirelessly to the reader. At its most basic, the chip will contain a serialised identifier, or licence plate number, that uniquely identifies that item, similar to the way many bar codes are used today.

There are three types of tags: active, passive and semi-passive.

Passive tags have no internal power source. These draw their power from the electromagnetic field generated by the RFID reader and then the microchip can send back information on the same wave. The reading range is limited when using passive tags.

Active transponders have their own transmitters and power source, usually in the form of a small battery. These remain in a low-power 'idle' state until they detect the presence of the RF field being sent by the reader. When the tag leaves the area of the reader, it again powers down to its idle state to conserve its battery. As a result, active tags can be detected at a

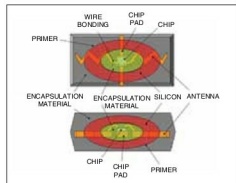


Fig. 3: Internal diagram of a typical RFID antenna

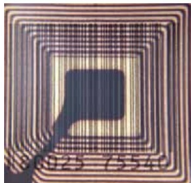


Fig. 4: Internal structure of a typical RFID tag

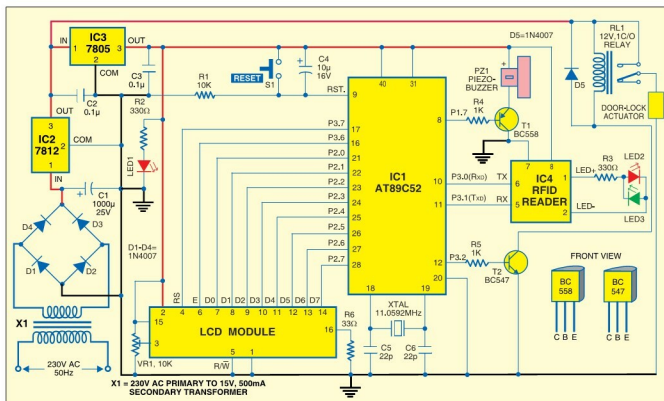


Fig. 5: Circuit of the RFID-based security system